

CVE-2025-53020 Proof of Concept

CVE-2025-53020 Proof of Concept - Gal Bar Nahum, Gal Bar Nahum.

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Content

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This reference is a source-code repository. The archive preserves its documentation at an exact commit; the code itself stays in a private mirror and is never checked out, built or run.

- Repository: <<https://github.com/galbarnahum/CVE-2025-53020-PoC>>
- Commit: `dd4bd9ca8a99375ad33a54be1b1648835289a6bb`
- Documents preserved: 1

README.md

Blob `5737a1b78c8d`, 3880 bytes, at commit `dd4bd9ca8a99`.

CVE-2025-53020 - Apache httpd HTTP/2 Memory Exhaustion PoC

A proof of concept demonstrating [CVE-2025-53020](#), a memory exhaustion vulnerability in Apache httpd's HTTP/2 implementation I found in mid-2025.

Deep dive: galbarnahum.com/posts/apache-httpd-cve-2025-53020

Vulnerability Overview

Apache httpd is vulnerable to a denial-of-service attack via HTTP/2 header processing. The vulnerability exists due to unnecessary memory duplication when handling repeated header names in HTTP/2 requests.

Attack Mechanism

1. **HPACK Exploitation:** The attack leverages HTTP/2's HPACK header compression by first sending an initial request to populate the server's HPACK dynamic table with a long header name
2. **Memory Amplification:** Subsequent requests reference this header name multiple times, but the server allocates new memory for each occurrence instead of reusing the existing entry
3. **Resource Exhaustion:** By sending batches of requests with thousands of header repetitions, an attacker can rapidly exhaust server memory

Impact

- Denial of Service (DoS)
- Server memory exhaustion
- Potential service unavailability

Requirements

- Python 3.7+
- `h2` - HTTP/2 protocol library
- `tqdm` - Progress bar for attack visualization

Installation

```
pip install h2 tqdm
```

Usage

Basic Usage

```
python poc.py --url https://target.example.com/
```

Command Line Arguments

Argument	Default	Description
<code>--url</code>	<i>required</i>	Target URL (e.g., <code>https://example.com/path</code>)
<code>--header-name-length</code>	4064	Length of the header name (max: 4064)
<code>--header-reps</code>	2063	Number of header repetitions per request (recommended max: 2063)
<code>--requests-per-batch</code>	1	Number of requests per batch
<code>--batches</code>	100	Number of batches to send

Argument	Default	Description
<code>--delay</code>	0	Delay between batches in seconds
<code>--num-headers-to-repeat</code>	1	Number of different header names to create (uses chars a, b, c, etc.)

Examples

Standard attack against HTTPS target:

```
python poc.py --url https://target.com/api/endpoint
```

Attack with custom header settings:

```
python poc.py --url http://localhost:8080/test --header-reps 1000 --batches 200
```

Attack with multiple distinct header names:

```
python poc.py --url https://target.com --num-headers-to-repeat 3 --header-name-length 3000
```

Slower attack with delay between batches:

```
python poc.py --url https://target.com --delay 0.5 --batches 500
```

Technical Details

HPACK Encoder Patch

The PoC includes a patch for the `h2` library's HPACK encoder to correctly handle empty header values. This fixes a bug where the encoder incorrectly indexes headers with empty string values (treating `""` as falsy instead of checking for `None`).

Constraints

- **Header name length:** Maximum 4064 bytes (limited by HTTP/2 frame constraints)
- **Header repetitions:** Recommended maximum 2063 per request
- **Multiple header names:** When using `--num-headers-to-repeat > 1`, header name length should be < 4000

Affected Versions

Apache httpd versions vulnerable to CVE-2025-53020. Check the official Apache security advisories for specific version information and patches.

Disclaimer

This tool is provided for educational and authorized security testing purposes only.

- Only use this tool against systems you own or have explicit written permission to test
- Unauthorized access to computer systems is illegal
- The authors are not responsible for any misuse or damage caused by this tool
- Always follow responsible disclosure practices

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